Simone Drago

Ph.D. Student in Information Technology

Milan, Italy simone.drago@polimi.it drago-simone Simone Drago

I am currently pursuing a Ph.D. at Politecnico di Milano, specializing in Reinforcement Learning from Human Feedback, with a focus on multi-objective decision-making under comparison feedback. I have authored five papers at top international conferences (ICML 2024-2026) and one journal article (AIJ 2025). I have industrial experience in addressing real-world problems by building formal models and developing theoretically grounded solutions, with applications spanning autonomous systems, gas network estimations, and corporate behavior prediction.

Education

Ph.D. **Politecnico di Milano**, Milan (IT), Information Technology — Sept. 2024 – Present

- **Research Focus:** Multi-Objective Reinforcement Learning from Human Feedback
- **Supervisor:** Prof. Alberto Maria Metelli

M.Sc. **Politecnico di Milano**, Milan (IT), Computer Science and Engineering — Sept. 2020 – May 2023

- **Final Grade:** 110/110 with Honors
- **Study Subjects:** Machine Learning, Reinforcement Learning, Recommender Systems, Data Science, Project Management
- **Master's Thesis:** Leveraging Machine Learning for UAV trajectory planning and performance estimation (supervisor: Prof. Marcello Restelli)
- **Academic Projects:**
  - Research on blockchain technology applied to ERP systems for B2B markets
  - Development of a top-K movie recommender system

Technical Skills

Programming Languages: Python, MATLAB, C, SQL, Java

Machine Learning / Scientific Computing Libraries: PyTorch, NumPy, SciPy, scikit-learn, pandas

Reinforcement Learning Libraries: Gymnasium, MO-Gymnasium, StableBaselines3

Tools: Git, LaTeX, Weights&Biases

Work Experience

Artificial Intelligence Research and Innovation Center, AI Researcher, Research Assistant (Assegno di Ricerca) at Politecnico di Milano — Milan, Italy; June 2023 – Sept. 2024; 1 year, 4 months

- **Autonomous Swarm Planning and Control**: Developed a simulation environment and state-of-the-art planning methods for autonomous coordination of a swarm of quadcopter drones, targeting real-hardware deployment.
- **Gas Network Distribution**: Built a machine learning pipeline to estimate gas distribution network topology from hundreds of thousands of residential and industrial consumption samples across northern Italy and optimized input node pressure to maximize network coverage while minimizing leak losses. Delivered a working POC that was integrated into the client's production tooling.
- **Company Behavior Prediction**: Designed a data cleaning and similarity-based classification pipeline to predict companies' future behavior given declaration data and official records, resulting in an MVP delivered to the client.

ML cube, AI Specialist, M.Sc. Internship — Milan, Italy; July 2022 – May 2023; 11 months

- **Autonomous Trajectory Planning**: Designed and implemented a full ML pipeline for autonomous UAV trajectory planning: generated a training dataset via a high-fidelity simulator, trained an MLP surrogate model to estimate key flight metrics (fuel consumption, altitude, travel time, trajectory length) at inference time, and integrated it into an automated planning procedure, reducing planning time from hours to seconds while producing trajectories qualitatively comparable to those of human experts.

Languages

Italian: Native speaker

English: Excellent – Cambridge English: Advanced, University of Cambridge, Grade A (CEFR Level C2), 2018

Publications

* Equal Contribution

International Conferences

[C5] **Simone Drago**, Marco Mussi, and Alberto Maria Metelli. *Online Compatible Reward Identification from Preference Feedback*. International Conference on Machine Learning (ICML), 2026. Acceptance Rate 26.6%.

[C4] **Simone Drago***, Marco Mussi*, and Alberto Maria Metelli. *Sleeping Reinforcement Learning*. International Conference on Machine Learning (ICML), 2025. Acceptance Rate 26.9%. (link: https://proceedings.mlr.press/v267/drago25a.html).

[C3] **Simone Drago**, Marco Mussi, and Alberto Maria Metelli. *Towards Theoretical Understanding of Sequential Decision Making with Preference Feedback*. International Conference on Machine Learning (ICML), 2025. Acceptance Rate 26.9%. (link: https://proceedings.mlr.press/v267/drago25b.html).

[C2] **Simone Drago**, Marco Mussi, and Alberto Maria Metelli. *Position: Constants are Critical in Regret Bounds for Reinforcement Learning*. International Conference on Machine Learning (ICML) – Position Paper Track, 2025. Acceptance Rate 19.7%. (link: https://proceedings.mlr.press/v267/drago25c.html).

[C1] Marco Mussi*, **Simone Drago***, Marcello Restelli, and Alberto Maria Metelli. *Factored-Reward Bandits with Intermediate Observations*. International Conference on Machine Learning (ICML), 2024. Acceptance Rate 27.5%. (link: https://proceedings.mlr.press/v235/mussi24a.html).

Journals

[J1] Marco Mussi, **Simone Drago**, Marcello Restelli, and Alberto Maria Metelli. *Factored-Reward Bandits with Intermediate Observations: Regret Minimization and Best Arm Identification*. Artificial Intelligence, 2025. (link: https://doi.org/10.1016/j.artint.2025.104362).

Workshops

[W2] **Simone Drago**, Marco Mussi, Leonardo Bianconi, and Alberto Maria Metelli. *Modeling Incomparability: A New Rationality Paradigm for Preference-Based Reinforcement Learning*. Workshop on Epistemic Intelligence in Machine Learning (EIML) at the International Conference on Machine Learning (ICML), 2026.

[W1] **Simone Drago** and Marco Mussi. *Open Problem: Tight Bounds for Kernelized Multi-Armed Bandits with Bernoulli Rewards*. Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET) at the International Conference on Machine Learning (ICML), 2024.

Academic Activities

Teaching Assistance

- Teaching assistant for the B.Sc. Course "Fondamenti di Informatica" at Politecnico di Milano (A.Y. 2026/27)
- Teaching assistant for the B.Sc. Course "Fondamenti di Informatica" at Politecnico di Milano (A.Y. 2025/26)
- Teaching assistant for the B.Sc. Course "Sistemi Informatici" at Politecnico di Milano (A.Y. 2024/25)

M.Sc. Theses Co-supervision

[1] Mattia Centrella - Online RLHF from Multi-Objective Feedback. Co-supervision. Supervisor: Prof. Alberto Maria Metelli (M.Sc. in Computer Science and Engineering, Ongoing)

[2] Leonardo Bianconi - “Multi-Dimensional Reward Learning from Preference Feedback”. Co-supervision. Supervisor: Prof. Alberto Maria Metelli (M.Sc. in Computer Science and Engineering, December 2025)

[3] Michele Simeone - “A Theory-Driven Approach to Large Language Models Alignment with Human Feedback”. Co-supervision. Supervisor: Prof. Alberto Maria Metelli (M.Sc. in Computer Science and Engineering, April 2025)

[4] Tomaso Castellani - “A novel approach for path planning with goal position uncertainty”. Co-supervision. Supervisor: Prof. Nicola Gatti (M.Sc. in Mathematical Engineering, April 2024)

Extracurricular Activities

Co-organizer of the Reinforcement Learning Summer School (2026): Co-organized RLSS 2026, a ten-day summer school with an attendance of ≈150 people from industry and academia, coordinating the selection, communication, and onboarding of international-level speakers from industry and academia.

Participation in Student Association (2021 – 2023): Participated as a team member at first and as a team manager later in the *PoliMi Data Scientists* student association of Politecnico di Milano. I organized the association's activities, managed the editorial content of a newsletter, and produced podcasts to promote AI and Data Science-related topics among university students.

First Ascent (2021): Selected among the top 20 Computer Science students in Italy for a Bending Spoons event.

Hackathon "services4migrants" (2017): Participated in a hackathon organized by the municipality of Milan, applying gamification concepts to support the access to public services by migrants within the city of Milan.

Awards

Top Reviewer at NeurIPS (2026): Nominated among top reviewers by Advances on Neural Information Processing Systems.

Gold Reviewer at ICML (2026): Recepient of "Best Reviewer" at International Conference on Machine Learning (Top 25% of Reviewers).

Young Researcher Best Publication Award (2025): Recipient of a prize for the best journal publications of the year, awarded by Politecnico di Milano.

Premio Migliori Matricole A.Y. 2017/18: Recipient of a scholarship for the best first-year students of academic year 2017/18, awarded by Politecnico di Milano.

B.Sc. and M.Sc. Student Scholarships: Reduction of tuition fees for merit (GPA > 27/30) — *each eligible year*.